Design Technology Co-Optimization for Digital Compute-in-memory with Area-efficient CFET 10T3P SRAM

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2026 IEEE International Symposium on Circuits and Systems (ISCAS) | 979-8-3315-7769-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISCAS66217.2026.11562810

***Abstract*—A high-throughput 10T3P SRAM using a complementary FET (CFET) design, with one more pFET read port (pRP) than the 8T2P SRAM for bit-parallel operations, can increase throughput and computing efficiency at the cost of only 1.1× cell footprint. The Digital Compute-in-memory (DCIM) macro and performance modeling, including the novel design of the adder tree for the CFET 10T3P SRAM, are comprehensively studied. Multi-V_T and dual rail technologies can significantly reduce SRAM read latency and short-circuit energy. Compared to the CFET 8T2P SRAM, the novel CFET 10T3P SRAM achieves 2.50X TOPS/W and 2.32X TOPS/mm² in DCIM applications.**

***Keywords*—Complementary Field Effect Transistor, Digital Compute-in-Memory, nanosheets, DTCO, Backside Power Delivery Network.**

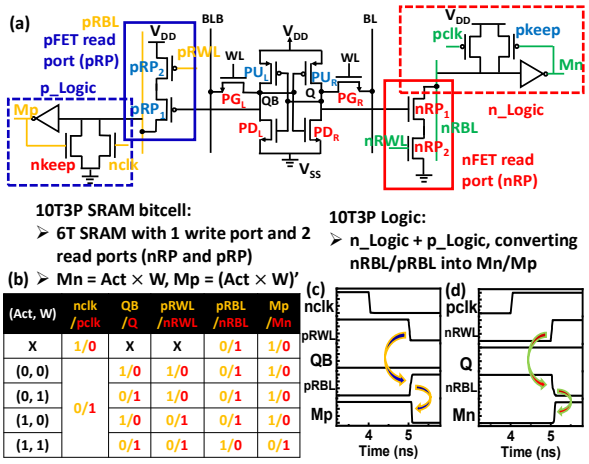
I. Introduction

The rapid advancement of AI increases the demand for high-performance computing. SRAM Digital Compute-in-memory (DCIM) offers energy and area efficiency without compromising accuracy [1-5]. To meet this demand, multi-port SRAMs supporting simultaneous read and write, such as 8T2P SRAM [2, 3] or 12T SRAM [4] using one-tier transistor technology, were reported for DCIM applications. Moreover, bit-parallel DCIM is another way to increase throughput at the cost of footprint [1], which can reduce the toggle rate in the adder tree and thus increase energy efficiency [6]. For advanced technology, monolithic complementary FET (CFET, 2-tier transistor technology) with common gate [7], split gate [8], heterogeneous channel materials [9], and backside power delivery network (BSPDN) [10] are reported to extend Moore's law. However, the CFET 8T2P (6N/2P) SRAM has low area efficiency due to the unbalanced nFET and pFET counts and the demand for significant strength in the nFET read port (nRP) [11, 12]. (**TABLE I**).

To implement bit-parallel multi-port SRAM DCIM using CFET, the 10T3P (6N/4P) SRAM is studied to improve area efficiency and throughput (**Fig. 1(a)**). Compared with CFET 8T2P SRAM, the CFET 10T3P SRAM has an additional pRP, offering 2× throughput than the CFET 8T2P SRAM with only a 10% footprint penalty. The corresponding n_Logic and p_Logic convert the analog signals on nRBL and pRBL into

SRAM Bitcell	Pros and Cons
6T [1]	• Fully explored in CFET (4-transistor footprints) • Low TOPS due to limitation on simultaneous read and write operations
8T2P [2, 3, 11, 12]	• Support simultaneous read and write operations ➔ high TOPS for AI-edge devices • Not balanced n/pFET counts (6N/2P for 8T2P) • Need a multi-fin read port to enhance readability ➔ At least 6-transistor footprints, low TOPS/mm²
10T3P (This work)	• Support simultaneous read and write operations • 1 more read port than 8T2P with the same footprint • pRP operation favors top nFET and bottom pFET placement in CFET • Bit-parallel MAC reduces the adder tree toggle rate ➔ TOPS ↑ with small area penalty, TOPS/mm² ↑ ➔ Small short circuit energy, TOPS/W ↑ • Need a pair of adder tree • Additional INV to handle the complementary outputs

TABLE. I. SRAM bitcell comparison using CFET technology for DCIM application.



(Act, W)	nclk /pclk	QB /Q	pRWL /nRWL	pRBL /nRBL	Mp /Mn
X	1/0	X	X	0/1	1/0
(0, 0)	0/1	1/0	1/0	0/1	1/0
(0, 1)		0/1	1/0	0/1	1/0
(1, 0)		1/0	0/1	0/1	1/0
(1, 1)		0/1	0/1	1/0	0/1

Fig. 1. (a) Schematics of 10T3P SRAM bitcell. (b) The truth table for each combination of controlling clock, activation (Act), and weight (W) in pFET read port (pRP) and nFET read port (nRP). Waveform for (Act, W) = (1, 1) in pRP (c) and nRP (d).

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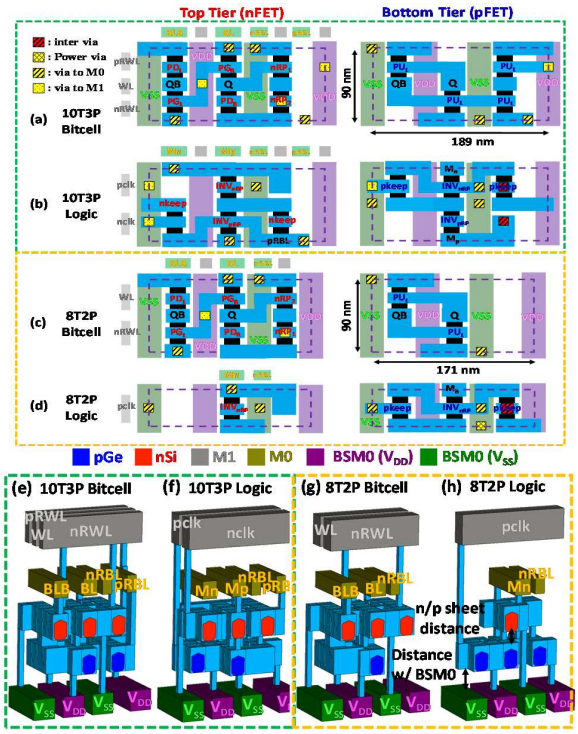


Fig. 2. CFET layout of top-tier nFET and bottom-tier pFET for 10T3P (a) SRAM bitcell, (b) Logic, and (c) (d) 8T2P references. (e-f) 3D structures of each part of the circuits.

digital signals, Mn and Mp, respectively. Mn represents the product of the activation (Act) and weight (W), i.e., Mn = Act × W, while Mp represents its complement, i.e., Mp = (Act × W)' (**Fig. 1(b)-(d)**) [2]. Since the nRP and pRP are pull-down and pull-up circuits, respectively, the multiply-accumulate (MAC) operations in nRP and pRP are performed with reversed waveform (**Fig. 1(c)(d)**). Design Technology Co-Optimization (DTCO) works like multi-V_T and dual rail technologies [6, 13] are used in this work to relieve the high-strength demand of transistors in the RP. The low BEOL coupling capacitance, the strong pFETs using Ge channel [14, 15], and the top nFET/bottom pFET placement [7] are used to improve MAC operations in the pRP, resulting in enhanced TOPS/W. After all, the DCIM performance matrices are benchmarked using Neurosim [16, 17].

II. Architectures of CFET 10T3P SRAM

A. Layouts and Device Parameters

The CFET 10T3P bitcell, integrating 6T SRAM and n/pRP, utilizes vertically stacked germanium pFET nanosheets (pNSs) and silicon nNSs with BSPDN. **Fig. 2(a)-(d)** and **Fig. 2(e)-(h)** show the layouts for each tier of 8T2P/10T3P SRAM bitcell/logic and their 3D structures, respectively. Note that the cell height of the 10T3P bitcell is 10% larger than that of the 8T2P bitcell due to the additional pRWL via. The corresponding Logic is structured with the same cell height as the SRAM bitcell (**Fig. 2(a)(b)**).

Fig. 3(a) lists all the device parameters and interconnect design rules [7, 15, 22, 23]. The parasitic resistances and capacitances are extracted, including the thin film resistivity, to analyze delay and power consumption in detail (**Fig. 3(b)**) [14, 15]. Copper is used as BSM0, M0, and M1 layers; the MOL metals are tungsten [24-27]. Note that 10T3P SRAM

(a)

Gate length	13nm
S/D length	20nm
Spacer length	6nm
Gate cut	12nm [22]
Gate extension	7.5nm [22]
S/D doping conc.	5×10^{20} cm^{-3}
Body doping conc.	10^{15}cm^{-3}
Body thickness	5nm
Suspension thickness	10 nm
NS floor# / EOT	3 / 0.86nm
n/p sheet distance	40nm [7]
Distance w/ BSM0	36nm [7]
Mobility ($cm^2V^{-1}s^{-1}$)	Si nNS: 300 [14] Ge pNS: 294 [14]
S/D Schottky Barrier Height (eV)	Si/TiSi$_2$:0.61 [30] Ge/NiGe:0.16 [31]

	Spacing	Thickness [23]
M0	9nm	27.5nm
M1/V0	14nm	52nm
BSM0	12nm	36nm

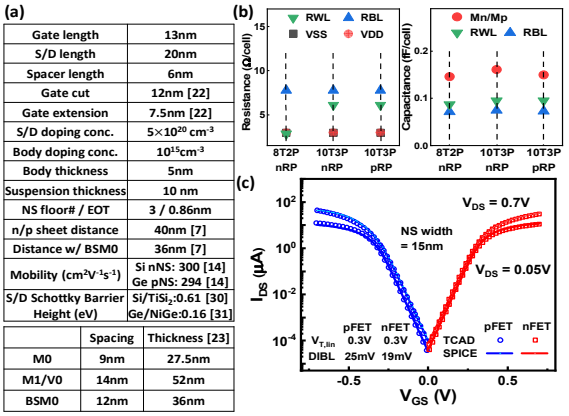
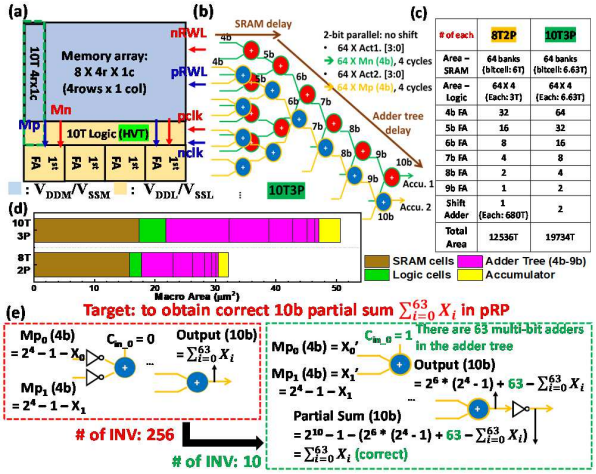


Fig. 3. (a) FEOL and BEOL parameters. (b) Resistance and capacitance per SRAM bitcell. (c) Transfer characteristics for Si nNSs and Ge pNSs.



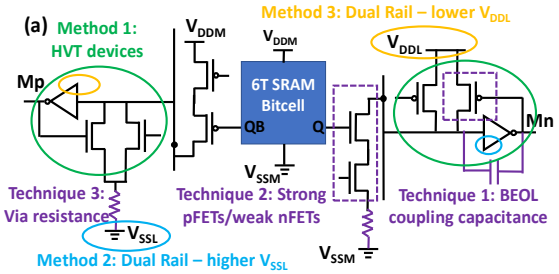
# of each	8T2P	10T3P
Area – SRAM	64 banks (bitcell: 6T)	64 banks (bitcell: 6.63T)
Area – Logic	64 X 4 (Each: 3T)	64X 4 (Each: 6.63T)
4b FA	32	64
5b FA	16	32
6b FA	8	16
7b FA	4	8
8b FA	2	4
9b FA	1	2
Shift Adder	1 (Each: 680T)	2
Total Area	12536T	19734T

Fig. 4. (a) 10T3P SRAM DCIM floor plan. (b) The adder tree structure with critical delay, and (c) total components for the 8T2P bit-serial and 10T3P bit-parallel SRAM DCIM macro. The area is represented in transistor size (T). (d) Area breakdown of SRAM DCIM macro. (e) The modification in the pRP adder tree to obtain the correct partial sum.

has a larger R_{WL} and parasitic capacitance than 8T2P due to its larger cell height and denser metal lines, respectively (**Fig. 3(b)**). The transfer characteristics of FEOL devices are obtained by TCAD, using mobility parameters calibrated with experimental data in our previous work [15] (**Fig. 3(c)**). It is observed that Ge pFETs exhibit 30% higher I_{ON} (V_{GS} = V_{DS} = 0.7V) compared to Si nFETs, which can be attributed to three primary factors: (1) increased gate capacitance [28], (2) enlarged scaling length and drain-induced barrier lowering (DIBL) [29], and (3) reduced S/D Schottky barrier height [30, 31]. The first two factors stem from the higher dielectric constant of Ge relative to Si, while the reduced Schottky barrier height is primarily due to Fermi-level pinning near the valence band of Ge. Additionally, since the DCIM performance metrics should include the adder tree [16], the BSIM-CMG model is calibrated with TCAD data to simulate the adder tree performance (**Fig. 3(c)**).

Based on the reported SRAM DCIM floor plan [18], the 10T3P SRAM DCIM floor plan (**Fig. 4(a)**) includes the distributed adder tree, distributed SRAM array, signal flow, dual rail scheme, and high V_T (HVT) device placement. Note that dual rail and HVT technologies relieve the high strength demand in RP transistors, combined with the CFET structure, CFET 10T3P shows a 53% area reduction compared to FinFET 8TSP SRAM [11].

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(b)	Higher V_{SSL}	Lower V_{DDL}
nRP	■ larger INV pull-up strength than V_{SSM} = V_{SSL} ■ Keeper V_G ↑, V_{SG} ↓ ➔ weak keeper	■ keeper V_S ↓, V_{SG} ↓ ➔ weak keeper
pRP	■ Keeper V_S ↑, V_{GS} ↓ ➔ weak keeper	■ larger INV pull-down strength than V_{DDM} = V_{DDL} ■ keeper V_G ↓, V_{GS} ↓ ➔ weak keeper

Fig. 5. (a) Three methods to enhance the readability for both the 8T2P and 10T3P. Three techniques in purple are utilized further to reduce read delay in 10T3P nRP and pRP. (b) Higher V_{DDL} and lower V_{SSL} on nRP/pRP to reduce read delay.

B. Delay Definition and Adder Tree Modifications

The critical delay and SRAM macro components (**Fig. 4(b)**) are used to simulate the DCIM performance based on the published simulators and models [16, 17]. One clock cycle includes SRAM read and adder tree delays for the critical delay. A DCIM macro with 64 banks x 4 rows x 4 columns of SRAM array is designed in this work. Therefore, a 6-stage adder tree is required (**Fig. 4(b)**). However, 10T3P SRAM CIM can output two pairs of MAC results in a clock cycle, which also involves a pair of adder trees to handle 2X output data and then achieve the "bit-parallel" MAC operations on DCIM [1, 6]. Besides the delay, the footprint of each component is estimated in terms of transistor size (T) to evaluate the area efficiency (**Fig. 4(c)(d)**). The 8T2P bitcell size is normalized as a 6T footprint. The 1-bit 28T full adder (FA) is used for the adder tree simulation and is expected to have a 16T footprint using CFET technology. The shift adders at the end of the adder tree are also included.

Since the Mp and Mn are in reversed waveforms (Mp = Mn', **Fig. 1(c)(d)**), a straightforward method to accumulate Mp correctly is to pass it through an additional INV before it enters the adder tree. However, an extra 256 INVs per adder tree would have to be used. In this work, a novel modification is applied to the adder tree by setting the carry-in (C_{in}) of the least significant bit (LSB) in each N-bit adder to "1". The correct partial sum is obtained after passing the output through a 10-bit INV in the adder tree output since all the additional '1's in the adder tree compensate for the difference (**Fig. 4(e)**). This modification is demonstrated to be functional in accumulating either unsigned or 2's complement data, and it avoids the demand for additional INV between each output bit and the 4-bit adder, reducing the INV number in the pRP adder tree from 256 to 10.

III. DTCO for Delay and Power Reductions

A. DTCO for SRAM Delay Reduction

Conventionally, 8T2P SRAM DCIM utilized either a weak keeper [2] or RP transistors with a large effective width

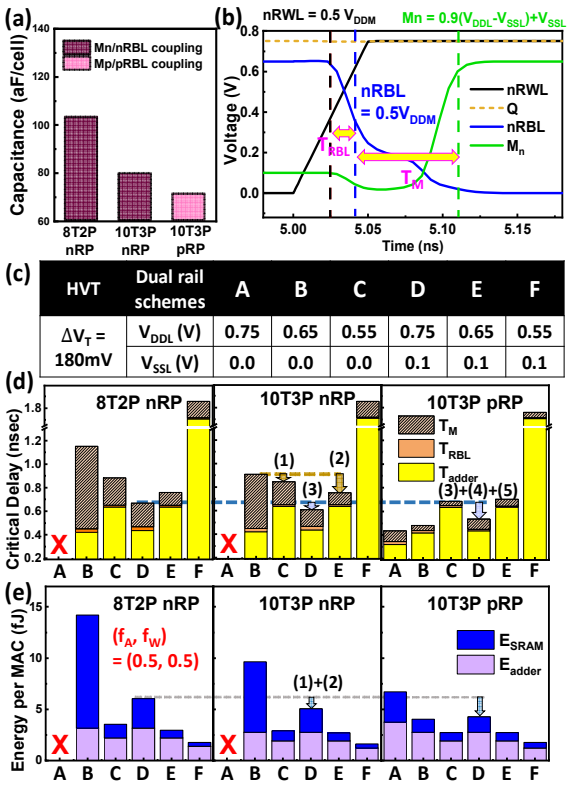


Fig. 6. (a) Mn/nRBL and Mp/pRBL coupling capacitance. (b) T_{RBL}/T_M definition for the read operation. (c) The six V_{DDL}/V_{SSL} schemes and the corresponding read delay in 8T2P and 10T3P SRAM. X means read failure. (d) Critical delay and (e) energy per MAC of SRAM DCIM under six V_{DDL}/V_{SSL} schemes.

[11] to boost read speed at the cost of additional footprints. Three methods are proposed in this work to reduce read delay without area cost: (1) HVT devices, (2) higher V_{SSL}, and (3) lower V_{DDL} (**Fig. 5(a)**).

Firstly, utilizing HVT devices in n_Logic and p_Logic can weaken the strength of both nkeep and pkeep to reduce the delay. Secondly, the dual rail scheme with lower V_{DDL} reduces the source voltage (V_S) of the pkeep and the gate voltage (V_G) of nkeep, and thus flips Mn and Mp rapidly, respectively. Thirdly, higher V_{SSL} can also help reduce read delay. Besides the keeper bias condition, the relative magnitude of V_{DDM}/V_{SSM} and V_{DDL}/V_{SSL} also affects the read delay. For pRP, the input gate voltage of the INV ranges from V_{SSL} to V_{DDM}. Thus, the INV has a significant pull-down strength when $V_{DDM}>V_{DDL}$, resulting in flipping the Mp rapidly. The individual impacts of V_{DDL} and V_{SSL} on nRP and pRP are summarized in **Fig. 5(b)**, and the smallest voltage range (V_{DDL} – V_{SSL}) is limited to 0.45V [5].

Moreover, three techniques in purple (**Fig. 5(a)**) are also utilized to further reduce read delay in 10T3P SRAM with the given V_{DDL}/V_{SSL} schemes: (1) reduction of BEOL coupling capacitance between nRBL (pRBL) and Mn (Mp). Since nRBL and Mn switch in different directions, decoupling these two signals can reduce read delay. In the 10T3P Logic, nRBL and Mn in nRP are decoupled by Mp, and so is the counterpart in pRP (**Fig. 2(e)**, **Fig. 6(a)**). (2) Slightly larger Ge pFETs I_{ON} relative to Si nFETs. (3) Series resistance due to the long via from top nFETs to BSPDN (VSS) degrades the nRP and nkeep strength, leading to decreases in the pRP delay at the cost of increasing the nRP delay. By utilizing these DTCO methods, the read delay in the CFET 10T3P SRAM is largely reduced.

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B. Critical Delay for MAC Operations

The waveform of Mn/Mp switching is divided into two parts to analyze (**Fig. 6(b)**): RBL delay (T_{RBL}) and Mn/Mp delay (T_M) are mainly controlled by the keeper and the inverter (INV) strength, respectively. Though increasing the ΔV_T can reduce T_{RBL}, the T_M may increase due to the weak INV strength. Note that the V_T tuning range in the GAA technology is limited to 200 mV [13]. Thus, the HVT devices with optimized V_T increment (ΔV_T) of 180 mV are selected for the lowest SRAM read delay. The loading capacitance of 0.72fF on Mn and Mp is used as the input capacitance of the 28T full adder [17].

Combining the adder tree delay simulated by SPICE (T_{adder}), the critical delay under all the dual rail schemes is listed (**Fig. 6(c)(d)**) [16]. Note that the individual effect of each method and technique in delay reduction is highlighted in **Fig. 6(d):** (1) lower V_{DDL}, (2) higher V_{SSL}, (3) lower Mn/nRBL (Mp/pRBL) coupling capacitance, (4) strong pFETs, and (5) via resistance. Under the same dual rail scheme, both nRP and pRP in 10T3P show a smaller read delay than nRP in 8T2P SRAM.

C. Energy Consumption for MAC Operations

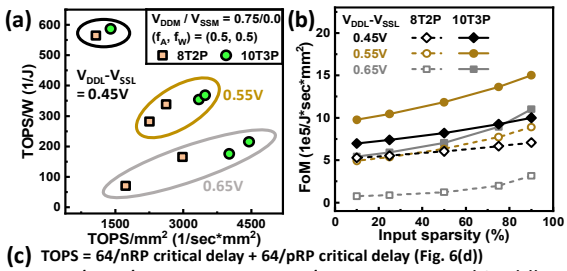
The total energy consumption has two components: SRAM read energy (E_{SRAM}) and adder tree energy (E_{adder}). E_{SRAM} includes dynamic, short-circuit, and leakage energy (E_{Dyn}, E_{SC}, and $E_{leakage}$). E_{adder} depends on the signal toggle rate inside the adder tree, while E_{SRAM} in the CIM macro depends on the fraction of bits equal to "1" in Act and W, denoted as f_A and f_W, respectively. The energy consumption for each bit-pair combination $(f_A, f_W) \in \{(1,1), (1,0), (0,1), (0,0)\}$ is denoted as $E(f_A, f_W)$. Therefore, the E_{SRAM} is a weighted summation as below (**Eq. 1**).

$$E_{SRAM}(f_A, f_W) = f_A \cdot f_W \cdot E(1,1) + (1-f_A) \cdot f_W \cdot E(0,1) + f_A \cdot (1-f_W) \cdot E(1,0) + (1-f_A) \cdot (1-f_W) \cdot E(0,0) \quad (1)$$

E(1, 1) is the main component inside the E_{SRAM} due to the large E_{SC} during the nRBL/pRBL flipping, and 10T3P pRP consumes much less E_{SC} than nRP due to a smaller read delay. E_{adder} of 8T2P and 10T3P are simulated by 1000 input patterns with toggle rates equal to 28% and 20%, respectively, considering the reported toggle rate reduction due to the bit-parallel DCIM operations [5]. Since the whole DCIM macro shares an adder tree, doing the 64 MACs in a clock cycle (Each MAC = 1b Act x 4b W), energy per MAC under (f_A, f_W) = (0.5, 0.5) is obtained by combining E_{SRAM} and E_{adder} (**Fig. 6(e)**). As a result, 10T3P reduces the energy per MAC as compared to 8T2P due to: (1) small SRAM read delay and thus smaller E_{SRAM}, and (2) smaller toggle rate and thus smaller E_{adder} (**Fig. 6(e)**).

IV. DCIM Performance Benchmark

The DCIM macro performance can be modeled by the critical delay and average power (**Fig. 6(d)(e)**). Since nRP and pRP in 10T3P have different clock signals, their critical delay and are calculated separately. **Fig. 7** compares the performance of the CFET DCIM macro, where the TOPS is defined as the number of MACs (64 for 8T2P and 64+64 for nRP+pRP in 10T3P) in a clock cycle divided by the critical delay of the corresponding read port. The TOPS/mm^2 doesn't include components outside the DCIM macro in this work. Note that under small voltage ranges, such as V_{DDL}/V_{SSL} =0.55/0.1V, the critical delay is significant and dominated by



(c) TOPS = 64/nRP critical delay + 64/pRP critical delay (Fig. 6(d))
TOPS/W = 1/nRP energy per MAC + 1/pRP energy per MAC (Fig. 6(e))
TOPS/mm^2 = TOPS/total area including adder tree (Fig. 4(c))

(f_A, f_W) =(0.5, 0.5)	N3 6T SRAM [16]	CFET 8T2P		CFET 10T3P (This work)	
Operation	Bit-serial	Bit-serial		2-Bit-parallel	
V_{DDL}/V_{SSL}	0.7/0.0	0.65/0.0	0.55/0.1	0.65/0.0	0.55/0.1
TOPS/W	330	70.4	564	176 (2.50X)	586 (1.03X)
TOPS/ mm^2	375	1729	1071	4014 (2.32X)	1340 (1.31X)
FoM	1.2e5	1.2e5	6.0e5	7.1e5 (5.80X)	8.2e5 (1.36X)

Fig. 7. (a) Benchmark for 8T2P and 10T3P SRAM in TOPS/W and TOPS/mm^2, under V_{DDM}/V_{SSM} = 0.75/0.0V and (f_A, f_W) = (0.5, 0.5). (d) FoM benchmark considering different input sparsity. (c) Formulas for 10T3P DCIM performance, corresponding figures, and the summary of CFET 8T2P and CFET 10T3P SRAM DCIM performance.

T_{adder}, while the energy per MAC is largely reduced, leading to high TOPS/W as an energy-efficient computing (black circle in **Fig. 7(a)**). On the other hand, large voltage ranges such as V_{DDL}/V_{SSL} = 0.65/0.0V will lead to area-efficient computing (gray circle in **Fig. 7(a)**).

The CFET 10T3P shows the enhancements of 2.50× TOPS/W and 2.32× TOPS/mm^2 with respect to CFET 8T2P. The improvement in 10T3P should be attributed to the small critical delay in 10T3P pRP and the almost same SRAM bitcell footprint for CFET 8T2P and CFET 10T3P SRAM (**Fig. 2(a)(c)**). Under the same V_{DDM}/V_{SSM}, the tradeoff between TOPS/W and TOPS/mm^2 can be tuned by V_{DDL}/V_{SSL}, giving designers flexibility to design with limited SRAM V_{min} [6] (**Fig. 7(a)**). The FoM [4], defined as TOPS/W × TOPS/mm^2, is benchmarked considering the varied input sparsity (**Fig. 7(b)**), where the input sparsity is defined as the fraction of bits equal to 0 [2]. The CFET 10T3P SRAM excels CFET 8T2P in FoM under all dual rail schemes and input sparsity levels, with the improvement varying from 1.36× to 5.80× under (f_A, f_W) = (0.5, 0.5) (**Fig. 7(c)**).

V. Conclusion

The CFET 10T3P SRAM DCIM macro, maintaining a similar 6-transistor footprint as CFET 8T2P SRAM, achieves up to 2.50× TOPS/W, 2.32× TOPS/mm^2, and 5.80× FoM with respect to the CFET 8T2P. The FoM of CFET 10T3P outperforms CFET 8T2P under all dual rail schemes and input sparsity scenarios. The nRP and pRP in 10T3P SRAM DCIM are improved by high V_{SSL} and low V_{DDL}, respectively. The geometry of CFET and BSPDN favors pRP MAC over nRP, making the 10T3P SRAM a compelling option for future high-performance computing with CFET DCIM architectures.

Acknowledgment

This work was supported in part by the National Science and Technology Council, Taiwan, under Grant NSTC 114-2218-E-002-033-MBK, NSTC 112-2923-E-002-013-MY3, NSTC 114-2634-F-A49-001, NSTC 112-2221-E-002-247-MY3, and PSMC, Taiwan.

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